

A Project Report on

House Price Prediction

Submitted to

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE**

In partial fulfillment of the requirements for the completion of

THIRD YEAR BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING

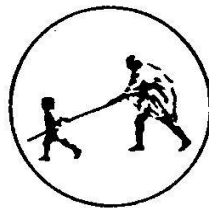
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Certificate



This is to certify that the report entitled

House Price Prediction

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in a satisfactory manner as a partial fulfillment of requirements for the
completion of Third Year in Computer Science and Engineering

to

MGM's College of Engineering, Nanded

Under

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
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House Price Prediction

Chapter 1: Introduction

House Price Prediction is one of the most important applications of Machine Learning in the real estate industry. Predicting house prices accurately helps buyers, sellers, real estate agents, and investors make better decisions. The price of a house depends on various factors such as location, number of bedrooms, bathrooms, living area, condition of the house, and other property features. Since real estate prices change frequently, manually estimating house prices can be difficult and time-consuming.

Machine Learning provides intelligent techniques for analyzing large amounts of housing data and predicting prices with high accuracy. Instead of relying only on human judgment, machine learning algorithms learn patterns from historical house sales data and use those patterns to estimate the prices of new houses.

In this project, different machine learning techniques are used to predict house prices. The dataset contains information about various houses, including their size, location, number of rooms, and selling prices. The model is trained using this data and then tested on unseen data to evaluate its performance.

The project also includes data preprocessing and visualization techniques. Data preprocessing helps remove unnecessary information and prepare the dataset for training. Visualization techniques help understand relationships between different house features and prices. Models such as Linear Regression and Gradient Boosting Regressor are implemented and compared based on their prediction accuracy.

The house price prediction system works by learning from previous housing records. After training, the model can predict the estimated selling price of a house based on its characteristics. This helps improve decision-making and reduces the chances of incorrect property valuation.

Objectives of the Project

1. To understand the concept of house price prediction using Machine Learning.
2. To learn how machine learning algorithms work for regression problems.
3. To implement Linear Regression and Gradient Boosting Regressor algorithms.

4. To preprocess and analyze housing datasets.
5. To evaluate the performance and accuracy of machine learning models.
6. To predict house prices based on property features.

Applications

- Real estate price estimation systems
- Property buying and selling platforms
- Real estate investment analysis
- Housing market research
- Online property recommendation systems
- Educational machine learning projects

This project is useful for real estate companies, property investors, home buyers, and students because it provides accurate house price predictions and helps in making informed real estate decisions. The implementation of machine learning techniques improves prediction accuracy and demonstrates the practical use of data science in the real estate sector.

Chapter 2: Literature Review and System Requirements

Many researchers and organizations have worked on house price prediction systems using different Machine Learning algorithms. Earlier systems relied on manual estimation methods where real estate agents and property experts predicted house prices based on their experience and market knowledge. These methods were often inaccurate because property prices depend on many factors and market conditions change frequently.

Modern house price prediction systems use Machine Learning techniques such as Linear Regression, Decision Trees, Random Forest, Support Vector Regression, and Gradient Boosting Regressor. These algorithms analyze historical housing data and identify patterns that influence property prices.

Linear Regression is one of the most commonly used algorithms for house price prediction because it is simple and easy to interpret. However, for more complex datasets, advanced algorithms such as Gradient Boosting Regressor provide better accuracy by combining multiple decision trees and reducing prediction errors.

The dataset used for house price prediction usually contains information such as the number of bedrooms, bathrooms, square footage, location, house condition, grade, and selling price. The quality of the dataset directly affects the performance of the prediction model. Therefore, data cleaning and preprocessing are essential steps in the project.

Hardware Requirements

1. Laptop or desktop computer
2. Minimum 4 GB RAM
3. Intel i3 or higher processor
4. Minimum 500 GB storage space

Software Requirements

1. Python programming language
2. Jupyter Notebook or VS Code
3. Machine Learning libraries such as Scikit-learn
4. Libraries such as Pandas, NumPy, Matplotlib, and Seabor

Existing System

Traditional house price estimation systems depend on real estate agents, market surveys, and manual calculations. These methods are often time-consuming and may produce inaccurate results because they depend heavily on human experience and judgment.

Proposed System

The proposed system uses Machine Learning algorithms to predict house prices automatically. The model is trained using historical housing data and learns the relationship between house features and property prices. After training, the system can estimate the price of a new house based on its characteristics.

Advantages of Machine Learning-Based House Price Prediction

- Automatic price prediction
- Faster decision-making
- Higher prediction accuracy
- Reduces human effort and errors

- Can analyze large housing datasets efficiently

Limitations

- Prediction accuracy depends on data quality
- Market fluctuations may affect results
- Missing or incorrect data can reduce performance
- Advanced models may require more computational resources

Despite these limitations, Machine Learning remains an effective approach for house price prediction and is widely used in the real estate industry for property valuation and market analysis.

Chapter 3: Methodology and Working Procedure

The methodology of this project explains the complete working process of the House Price Prediction System. The first step is collecting the dataset. The dataset contains information about houses and their selling prices. The dataset includes features such as bedrooms, bathrooms, living area, floors, location, condition, grade, and price. The dataset is usually stored in CSV format.

After collecting the dataset, preprocessing is performed. Preprocessing is very important because raw housing data may contain unnecessary columns, missing values, and inconsistent data.

Steps in Data Preprocessing

1. Data Loading

The housing dataset is loaded into Python using the Pandas library for further analysis and processing.

2. Data Cleaning

Unnecessary columns such as house ID are removed because they do not contribute to price prediction.

3. Handling Missing Values

Missing or null values are identified and handled to improve data quality and model performance.

4. Feature Selection

Important features such as bedrooms, bathrooms, square footage, condition, grade, latitude, and longitude are selected for training the model.

5. Data Visualization

Graphs and charts are created to understand the relationship between house features and prices. Scatter plots and bar charts help identify trends in the dataset.

After preprocessing, the dataset is prepared for machine learning. Since machine learning models work with numerical data, all selected features are arranged into input variables, and house price is selected as the target variable.

The dataset is then divided into training and testing data. Usually, 90 percent data is used for training and 10 percent for testing.

The machine learning models are trained using the training dataset. In this project, Linear Regression and Gradient Boosting Regressor are used. During training, the models learn the relationship between house features and house prices. After training, testing is performed using unseen housing data.

Performance Metrics

1. Accuracy Score
2. Mean Absolute Error (MAE)
3. Mean Squared Error (MSE)
4. R^2 Score (Coefficient of Determination)

Backend Working Process

1. Housing dataset is loaded.

2. Data preprocessing is applied.
3. Features are selected.
4. Training and testing datasets are created.
5. Machine learning models are trained.
6. House price prediction is generated.
7. Results are evaluated and displayed.

Frontend

This project is mainly implemented using Jupyter Notebook for data analysis, visualization, model training, and prediction. Various graphs and prediction outputs are displayed to the user for analysis.

This methodology makes the project simple, efficient, and easy to understand for students. It demonstrates how Machine Learning techniques can be used to predict house prices accurately using historical housing data.

Chapter 4: Results, Advantages, Limitations and Conclusion

The House Price Prediction project produced good results using Linear Regression and Gradient Boosting Regressor algorithms. After training and testing, the models were able to predict house prices with good accuracy. Among the two models, Gradient Boosting Regressor achieved the highest accuracy of approximately 91.94%, while Linear Regression achieved around 73% accuracy.

The accuracy of the model depends on factors such as dataset quality, feature selection, preprocessing techniques, and the machine learning algorithm used. Proper data cleaning and preprocessing significantly improve prediction performance.

Results

1. Housing data was successfully analyzed and visualized.
2. Machine learning models were trained using historical housing data.
3. House prices were predicted based on property features.
4. Gradient Boosting Regressor achieved better accuracy than Linear Regression.
5. The system provided reliable house price estimations.

Advantages of the Project

1. Fast and automatic price prediction.
2. Reduces manual effort in property valuation.
3. Provides high prediction accuracy.
4. Helps buyers and sellers make informed decisions.
5. Can analyze large housing datasets efficiently.
6. Useful for educational and research purposes.
7. Demonstrates practical application of Machine Learning.

Limitations of the Project

1. Prediction accuracy depends on data quality.
2. Missing or incorrect data can affect results.
3. Sudden market changes may reduce prediction accuracy.
4. Limited to the features available in the dataset.
5. Advanced models may require higher computational resources.

Future Scope

1. Use Deep Learning algorithms for improved accuracy.
2. Integrate real-time real estate market data.
3. Develop a web-based user interface for easy access.
4. Use larger and more diverse datasets.
5. Add location-based and market trend analysis.
6. Integrate with real estate websites and applications.
7. Implement mobile application support.

Conclusion

House Price Prediction using Machine Learning is an effective and practical project. The system automatically predicts property prices based on house characteristics such as location, bedrooms, bathrooms, living area, and condition. The project demonstrates the practical application of Machine Learning and Data Science in the real estate industry.

The project also helps students understand important concepts such as data preprocessing, feature selection, data visualization, model training, testing, and prediction. The implementation of Linear Regression and Gradient Boosting Regressor shows how different

machine learning algorithms can be used to solve real-world regression problems.

Overall, this project is useful, educational, and practical for real estate applications. It provides accurate house price predictions and can be further enhanced using advanced Machine Learning, Deep Learning, and Artificial Intelligence techniques.